



Two-dimensional liquid chromatography of polystyrene–polyethylene oxide block copolymers

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ABSTRACT

In this study, liquid chromatography at critical conditions of polystyrene (PS) and polyethylene oxide (PEO) is used as the first dimension for the two-dimensional analysis of PS-*b*-PEO copolymers. Comprehensive two-dimensional liquid chromatography, with size exclusion chromatography as the second dimension, reveals information about the molar mass distributions of all separated fractions from the first dimension. Furthermore, fractions eluting at the critical conditions of one block were collected and subjected to analysis in the second dimension at the critical conditions of the other block. These fractions were analysed by FTIR to determine their chemical compositions. The combination of the above approaches and the calibration of the evaporative light scattering (ELS) detector for the first-dimensional analysis yield deep insights into the molecular heterogeneity of the block copolymer samples. The composition of the samples and the chemical composition of the real block copolymer are also calculated by combining results obtained at both critical conditions.

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1. Introduction

Amphiphilic copolymers generally consist of covalently bonded hydrophilic and hydrophobic blocks. Due to the incompatibility of these blocks with each other and their different affinities for different solvents, they exhibit surface activity. Recently, these materials have attracted considerable attention due to their applications in several different fields: for the modification of surfaces to control protein adsorption, nano- and biomaterials, controlled release, nanofiltration membranes, surface modification of biomaterials, synthetic lubricants, emulsifiers, and for biomedical applications [1–4].

In most cases polyoxyethylene is the hydrophilic block, while alkyl, aryl alkyl, poly(ether), poly(ester), poly(styrene), etc. are the hydrophobic blocks. The block copolymers consisting of ethylene oxide and styrene are amphiphilic in nature. The amphiphilic nature of PS-*b*-PEO has been exploited for many different industrial applications such as polymeric surfactants in the oil, pharmaceutical, agriculture, paper and detergent industries, as well as compatibilisers in polymer blending, dispersions, stabilisers, and templates for the preparation of inorganic nanoparticles [5]. Exploiting the self-assembly of PS-*b*-PEO and PS-*b*-PEO-*b*-PS to

ordered nanostructures, novel applications such as the preparation of mesoporous silica films with different pore sizes [6] and electrolytes for rechargeable batteries have been developed [7]. Darling demonstrated in a review that the size and shape of self-assembled amphiphilic block copolymers are tuneable through the synthetic chemistry of the constituent molecules [8].

During the synthesis of PS-*b*-PEO copolymers, side reactions can take place leading to the formation of homopolymers with different functionalities of both types of monomers involved, along with the targeted block copolymers. The presence of homopolymers and the size of the different blocks have a significant effect on the final properties of these materials [8]. Therefore, characterisation of these block copolymers is vital to develop the structure–property relationships and to improve synthesis procedures. The application of spectroscopic methods such as NMR, MS, and FTIR to the bulk products only yields the average composition of the sample, which is not always conclusive with regard to information about all the components. These methods are, however, very often informative when used after fractionation by HPLC or other fractionation techniques.

Liquid chromatography is one of the most powerful methods for the characterisation of the molecular heterogeneity of complex polymers and liquid chromatography at critical conditions (LCCC) has been employed successfully for the fractionation of many different types of polymer blends, block copolymers, and polymers with different functionalities, etc. LCCC is one of the major modes of liquid chromatography employed as the first dimension in multidimensional liquid chromatography [9–31].

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In this study, liquid chromatography at the critical conditions of PS and PEO is used to separate the block copolymers from respective homopolymers. In one of our previous papers, the critical conditions of PS were established using a tetrahydrofuran (THF)–water mobile phase. However, due to the limited solubility of the polymers being investigated, this system only works for chains with a molar mass of up to 10 kg/mol of each block [32]. In this study we used a different mobile phase system in order to extend the molar mass range. This approach should provide information on the molar mass distribution of the non-critical block in the block copolymer and allows separation of critical homopolymers from the rest of the sample if there are any. An ELS detector is calibrated under the above conditions and the homopolymer content is quantified. Comprehensive two-dimensional liquid chromatography is used to obtain the molar mass distribution of the fractions separated at the critical conditions of both PS and PEO. As a next step, fractions are collected and subjected to further analysis at the critical conditions of the other block, as well as by FTIR for qualitative as well as quantitative analysis. The study is further extended by combining the results obtained at both critical conditions to quantify the homopolymer content as well as the chemical composition distribution of the real block copolymers. Recently, this approach has been demonstrated for PDMS-*b*-PS copolymers [33,34] to obtain valuable information regarding the complex polymer system. In this study, we applied a similar approach to other commercially important PS-*b*-PEO copolymers.

2. HPLC of polymers

The separation mechanism in liquid chromatography of polymers depends on the size/pore size ratio and on the so-called interaction parameter c , which describes the interaction of the structural unit with the stationary phase [35]. This parameter is negative in size exclusion chromatography (SEC), in which retention decreases with an increase in molar mass, and is positive in liquid adsorption chromatography (LAC) where retention increases exponentially with the number of repeat units.

In both cases, the elution volume V_e depends upon the distribution coefficient K of the polymer between the interstitial volume V_i and the pore volume V_p :

$$V_e = V_i + K \cdot V_p \quad (1)$$

This equation is well known in SEC (where K can assume values between 0 and 1): a polymer of very high molar mass (above the exclusion limit) elutes at V_i (as $K=0$), while small molecules (which have access to the entire pore volume) elute at the void volume $V_0 = V_i + V_p$ (as $K=1$).

In LAC, the distribution coefficient can assume very high values (much larger than 1). The distribution coefficient is related to the free energy change related to the transition of the polymer from the interstitial volume into the pores of the stationary phase:

$$\ln K = \Delta G = \Delta H - T \cdot \Delta S \quad (2)$$

In general, the driving force in SEC is the entropy change, while in LAC retention is governed by the enthalpic interaction of the repeat unit with the stationary phase.

For a given system (polymer–stationary phase–mobile phase) there may be a mobile phase composition (and temperature), where entropic and enthalpic contributions compensate each other ($\Delta H = T \Delta S$), hence $\Delta G = 0$ and $K = 1$.

The transition point between SEC and LAC is often termed as the “critical conditions” or the critical adsorption point (CAP). At the CAP, the interaction parameter has a value of zero: under such conditions, a non-functional homopolymer elutes – independent of its molar mass – at the void volume of the column. In other words,

the polymer chain becomes chromatographically invisible [36–39], hence it is possible to separate a complex polymer according to other structural units (different from the repeat unit), such as end groups or other blocks.

The selection of suitable column, mobile phase, and temperature is very important for establishing critical conditions for a particular type of polymer. The critical conditions at a given temperature are characteristic of polymer–adsorbant–eluent system. The ideal critical conditions for high polymers are those, which result in size exclusion elution of the non-critical components. Generally, critical conditions for nonpolar polymers can be established on a RP column in low mobile phase polarity. Under these conditions, polymers which are more polar than the critical ones elute in the size exclusion regime. Similarly, in high mobile phase polarity on a NP column, critical conditions for polar polymers can be established and nonpolar polymers elute in the size exclusion regime under these conditions [17]. The classic way of establishment of critical conditions is to select carefully column, mobile phase, and temperature for a particular polymer type and perform some isocratic runs of homopolymer standards in different mobile phase compositions. Of course this approach requires many isocratic runs to reach the critical point. Bashir et al., proposed a strategy for the fast determination of the CAP by gradient chromatography. For high molar mass polymers, the composition at elution is close to the critical composition [40]. This approach leads to considerable reduction in the number of experiments for establishing critical conditions.

These three modes of liquid chromatography can be combined in the analysis of functional polymers and copolymers by two-dimensional chromatography.

In the case of block copolymers the interaction parameters c_A and c_B of the individual structural units (A and B) may have different values. In a mobile phase, where both c_A and c_B are strongly negative, the molar mass distribution (MMD) can be determined by SEC, with retention decreasing with increasing molar mass. In SEC, the influence of chemical composition is often rather small. SEC yields the MMD of the entire molecule (provided that the calibration functions for A and B are comparable).

SEC is, however, not capable of discriminating mixtures of homopolymers with different repeat unit or polymers with the same repeat unit, but different functionality. **This can be achieved by LCCC**

At the CAP of the repeat unit A, the elution volume is independent of the chain length of block A. In this case, AB copolymers can be separated according to block B.

Basically, there are two possible situations, which may be utilised:

1. If the interaction parameter c_B of block B is negative, the block copolymer will elute earlier than the homopolymer A (in SEC order). Obviously, situation 1 will only be useful if block B is sufficiently large: it can be used for the separation of homopolymers from block copolymers [23,24,37,41–44].
2. If c_B is positive, the block copolymer elutes later than the homopolymer A (in LAC order) [16,45]. Situation 2 is limited to short blocks in the case of strong interaction (i.e. large c_B). It can be very useful, if the non-critical block is monodisperse [44].

3. Experimental

3.1. Synthesis of PS-*b*-PEO diblock copolymers

PS-*b*-PEO diblock copolymers were produced by Polymer Standards Service GmbH (Mainz, Germany) by sequential living anionic polymerisation of styrene and ethylene oxide [46]. Briefly, styrene was polymerised at -78°C in THF using cumylpotassium as

Table 1
Block copolymer samples information according to manufacturer.

PS-b-PEO #	Mw (g/mol)			Chemical composition	
	Total	PS	PEO	PS (%)	PEO (%)
1	–	1500	3170	32	68
2	–	1500	3960	28	72
3	7090	3940	3150	56	44
4	31,930	2930	29,000	9	91
5	60,000	30,000	30,000	50	50
6	91,500	30,000	61,500	33	67
7	134,000	30,000	104,000	22	78
8	218,000	109,000	109,000	50	50

initiator. After 1 h, an aliquot was taken to recover the polystyrene precursor for monitoring purposes. Ethylene oxide (EO) was then cryodistilled into the reactor. At this stage, the orange colour of the solution disappeared and a pale yellowish colour appeared. After raising the temperature to 40 °C, the reaction was continued for 2 days to complete the polymerisation of the second monomer. The polymerisation was terminated by the addition of methanol. The PS-b-PEO diblock copolymer formed was isolated by precipitation in hexane, followed by careful drying by dissolution in purified benzene. The solvent was removed through three freeze–thaw cycles in a high-vacuum line. The information about the block copolymers as provided by the manufacturer is given in Table 1.

3.2. Liquid chromatography

PS and PEO standards were purchased from Polymer Laboratories (Church Stretton, Shropshire, UK) and Polymer Standards Service (Mainz, Germany), respectively. The HPLC solvents; N,N-dimethylformamide (DMF) and THF were purchased from Sigma–Aldrich.

For LCCC, the mobile phase was delivered by a Waters Alliance 2690 separation module (comprising of a solvent mixer, vacuum degasser, and autosampler with a 100 µL sample loop) at a flow rate of 0.5 mL/min. The system was connected to an Agilent 1100 series variable wavelength UV detector (set at 254 nm) and a PL-ELS 1000, with data being recorded by PSS WinGPC software (version 7). Nitrogen was used as the carrier gas in the evaporative light scattering detector (ELSD) at a flow rate of 1.5 L/min. Evaporator and nebuliser temperatures were set at 180 °C and 80 °C, respectively. Mobile phases were mixed by volume. All columns used were placed in a temperature controlled column oven, maintained at 30 °C for all measurements. Sample concentrations were between 3.0 and 10.0 g/L.

The following columns were used for LCCC in this study:

- (1) **Symmetry 300** (Waters, Ireland): silica-based octadecyl phase, 250 mm × 4.6 mm, particle diameter 5 µm, pore size 300 Å, surface area 113 m²/g, carbon load 8.35%.
- (2) **Nucleosil Si 300** (Macherey–Nagel, Germany): plain silica, 250 mm × 4.6 mm, particle diameter 5 µm, pore size 300 Å.

The fractions, collected manually at the critical conditions of one block, were subjected to further analysis by LCCC at the critical conditions of the other block, as well as FTIR.

3.3. FTIR analyses

For qualitative studies, FTIR spectra were recorded in reflectance mode on a Thermo Nicolet iN10 spectrometer (Thermo Scientific, Waltham, MA) using a cooled MCT detector. Spectra were recorded at 4 cm^{−1} resolution with 64 scans being recorded for each spectrum. The samples were prepared by depositing a droplet of the sample in solution (DMF) onto a reflective gold surface. The

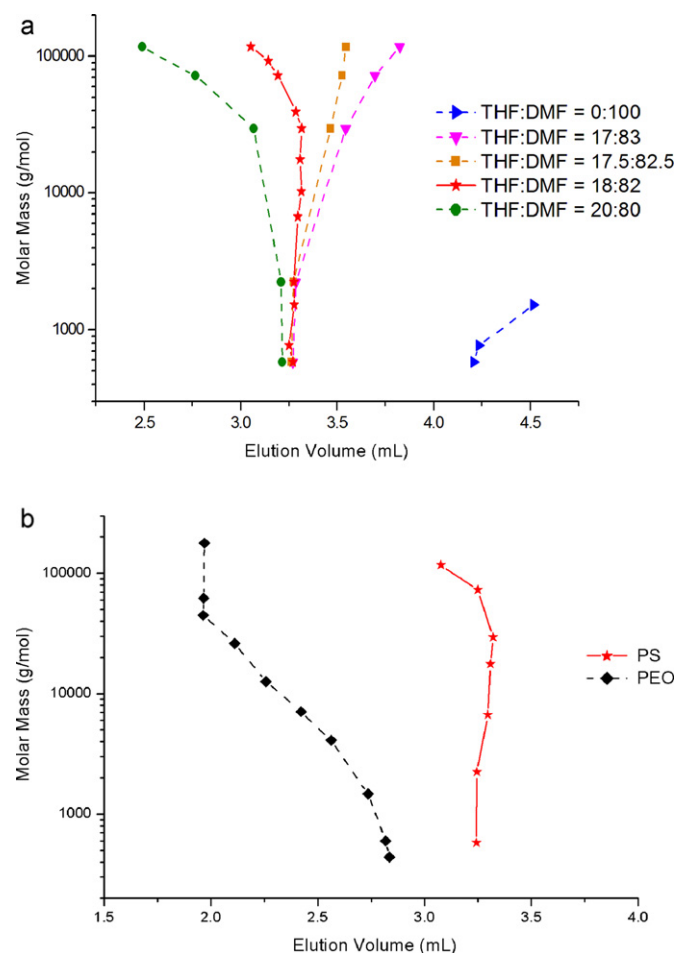


Fig. 1. (a) CAP of PS on a Symmetry RP octadecyl 300 Å column in THF-DMF and (b) elution behaviour of PEOs at the CAP of PS on a Symmetry RP octadecyl 300 Å column in THF-DMF (18–82).

solvent was then evaporated prior to analysis. Multiple spectra were recorded for each sample to ensure consistent results.

For quantification, FTIR spectra were recorded in transmission mode on a Thermo Nicolet iS10 spectrometer (Thermo Scientific, Waltham, MA) equipped with a demountable solution cell. Spectra were recorded at 4 cm^{−1} resolution with 64 scans recorded for each spectrum. All samples were analysed in solution (DMF, dried) at a fixed concentration of 5 mg/mL utilising a path length of 0.025 mm and ZnSe windows. The vibration modes associated with PS (monosubstituted aromatic out-of-plane C–H bending at 700 cm^{−1}) and PEO (C–O–C asymmetric stretch at 1145 cm^{−1}) were used for quantitative analysis. A calibration curve was constructed for quantification of the copolymer composition using PS and PEO standards. The calibration was fitted with a 3rd order polynomial equation. The coefficient of determination of the fit was 0.9903.

3.4. Two-dimensional liquid chromatography

For two-dimensional liquid chromatography, sample fractions from the first dimension were transferred to the second-dimension column via an electronically controlled eight-port valve system (VICI Valco instruments, Houston, TX, USA), equipped with two 100 µL sample loops. The second dimension consisted of a waters HPLC 515 pump and a 50 mm × 20 mm PSS GRAM column. Detection in the second dimension was accomplished by using an ELSD. Nitrogen was used as carrier gas in the ELSD at a flow rate

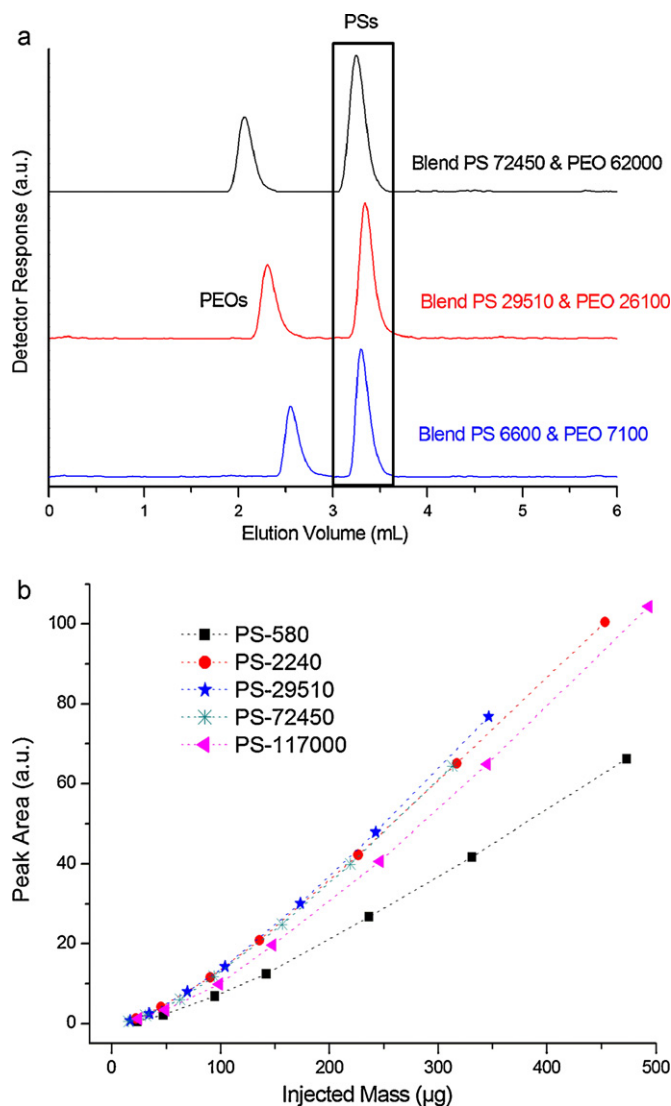


Fig. 2. (a) Elugrams of blends of PS and PEO calibration standards at the CAP of PS on a Symmetry RP octadecyl 300 Å column in THF–DMF (18–82) and (b) ELSD calibration curves for PS at its critical conditions (THF–DMF: 18–82) on a Symmetry RP octadecyl 300 Å column.

of 1.5 L/min. Evaporator and nebuliser temperatures were set at 230 °C and 130 °C, respectively. The flow rates used in the first and second dimensions were 0.02 mL/min and 2.5 mL/min, respectively. Sample concentrations were between 15.0 and 20.0 g/L. DMF was used as the solvent for the second dimension analysis. Data were recorded using PSS WinGPC software (version 7).

4. Results and discussion

4.1. Separations at the CAP of PS

Liquid chromatography at the critical conditions of PS allows the molar mass independent elution of PS homopolymers and the separation of the PS-*b*-PEO block copolymers which is governed by the PEO block irrespective of the size of the PS block. These conditions are very useful for the analysis of complex polymer mixtures. In our previous study under critical conditions of PS for PS-*b*-PEO copolymers, only low molar mass samples could be analysed [32]. In the current study, the critical conditions for PS were established using a different solvent combination in order to extend the molar mass range. The CAP of PS was realised in a DMF–THF mobile phase

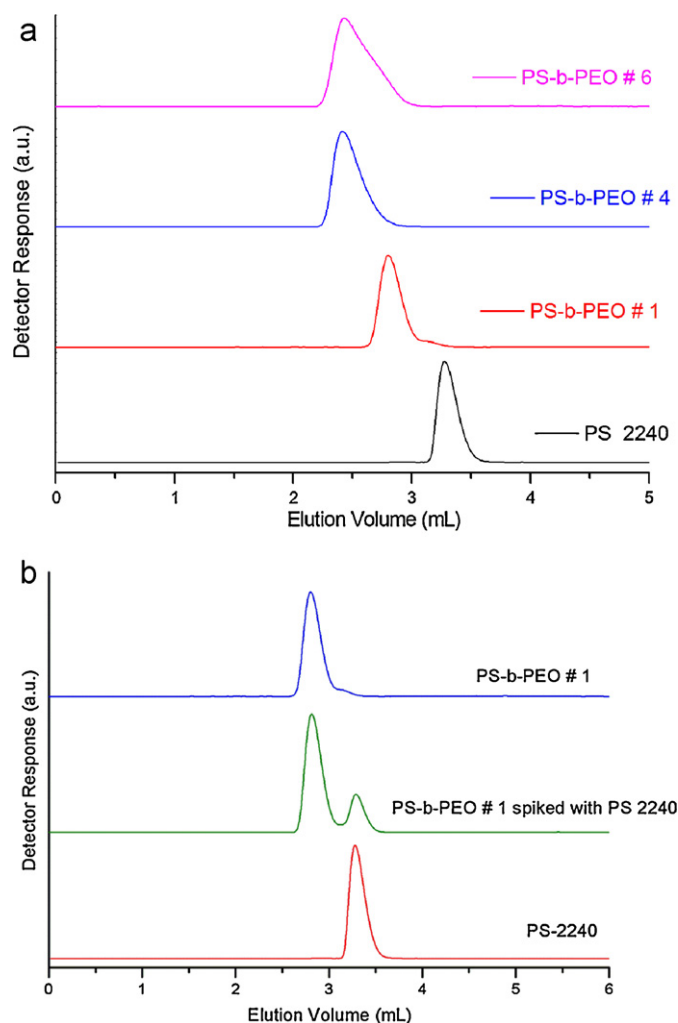


Fig. 3. (a) Elugrams of PS-*b*-PEO copolymers at the CAP of PS on a Symmetry RP octadecyl 300 Å column in THF–DMF (18–82) and (b) overlay of PS-*b*-PEO #1 and the same sample spiked with a PS standard at the CAP of PS on a Symmetry RP octadecyl 300 Å column in THF–DMF (18–82).

on a RP octadecyl column [14,47]. This allowed us to extend the molar mass range up to the hundred thousands. As demonstrated in Fig. 1a, PS calibration standards adsorb in 100% DMF on a silica-based octadecyl column. The higher molar mass standards elute later than the lower molar mass standards.

The elution volume decreases as the amount of THF in the mobile phase is increased. A point is reached at which all PS standards elute at more or less the same elution volume irrespective of their molar masses (THF–DMF: 18–82). These conditions are called the critical conditions or the CAP of PS for the specific column, temperature and mobile phase conditions used. If the amount of THF is increased further, PS becomes excluded from the pores of the stationary phase, resulting in the higher molar mass standards eluting earlier than the lower molar mass standards. Under the critical conditions for PS, the PEOs have no enthalpic interaction with the stationary phase and elute in size exclusion mode (Fig. 1b). The SEC calibration curve for PEOs under the above conditions is not ideal as evidenced by large differences in the elution volumes for even lower molar mass standards (PSs and PEOs), which in principle should not elute too far from each other [9]. Furthermore, the PEO polymer chains become completely excluded from the pores of stationary phase under these conditions at molar masses above 40 kg/mol. This renders inaccurate calculation of the molar mass distribution of the non-critical block. The two components of the block copolymers

being investigated have distinctly different polarities, which might be the reason for this strange behaviour. Consequently, the hydrodynamic volume of the PEOs, under the conditions used, is larger than one would ideally expect. We have also noticed that the hydrodynamic volumes of PSs and PEOs in pure DMF are different from each other, resulting in the characteristic SEC calibration curves for the second-dimension analysis (discussed later, Fig. 6). In our previous study [32], we found a similar type of behaviour for PEOs at the critical conditions of PS in the THF–water mobile phase system. Nonetheless, the PS homopolymers are successfully separated from the rest of the sample.

Fig. 2a depicts the elugrams of blends of PS and PEO calibration standards at the critical conditions of PS illustrating the separation obtained. It can be seen that PSs elute independent of their molar masses at more or less the same elution volume of approximately 3.3 mL, while PEOs show molar mass dependence and elute in size exclusion mode earlier than the PS standards. The ELS detector is calibrated for PS by injecting varying masses of different PS molar mass standards. The calibration curves for different molar mass PS standards are shown in Fig. 2b.

These calibration curves are used for quantification of PS homopolymers in the sample. There can be several factors affecting the calibration curves. For low molar mass standards, low oligomers might be evaporated with the mobile phase under the stringent evaporating conditions leading to smaller peak areas compared to some of the high molar mass standards [48]. For higher molar mass standards, limited recovery of polymers could be a factor, namely the incomplete elution of some of the injected polymers [11,49]. Another important factor which results in smaller peak areas for higher molar mass standards could be a change in droplet formation with increasing molar mass. It can be seen from Fig. 2b that PS standards smaller than 2.2 kg/mol and larger than 72.5 kg/mol have decreased peak areas compared to the other standards. The slightly smaller peak areas for lower molar mass standards can be attributed to the first reason, while for higher molar mass standards, smaller peak areas can be explained by the latter reasons.

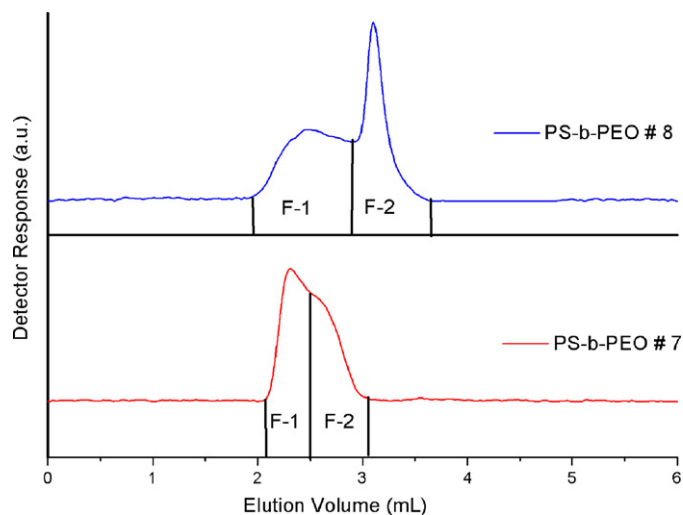


Fig. 4. Fractionation of PS-b-PEO copolymers at the CAP of PS on a Symmetry RP octadecyl 300 Å column in THF–DMF (18–82).

LCCC of PS has been used to investigate PS-b-PEO copolymers. The elution of the block copolymer should be governed by the PEO block, while the conditions allow the separation of PS homopolymers if there are any in the sample. Fig. 3a illustrates the elugrams of some of the block copolymers being investigated. It can be seen that PS calibration standards elute at 3.30 mL and, therefore, any PS homopolymers present in the block copolymers would also elute at this elution volume, which is obviously not the case for the samples shown in Fig. 3a, with the exception of PS-b-PEO # 1 which showed a slight shoulder around the critical elution volume.

The block copolymers are separated according to the molar mass distribution of PEO. As explained earlier, the calculation of the molar mass distribution of the PEO block under these conditions is not accurate enough to reach a useful conclusion. On the other hand, these conditions allow the separation of PS homopolymers

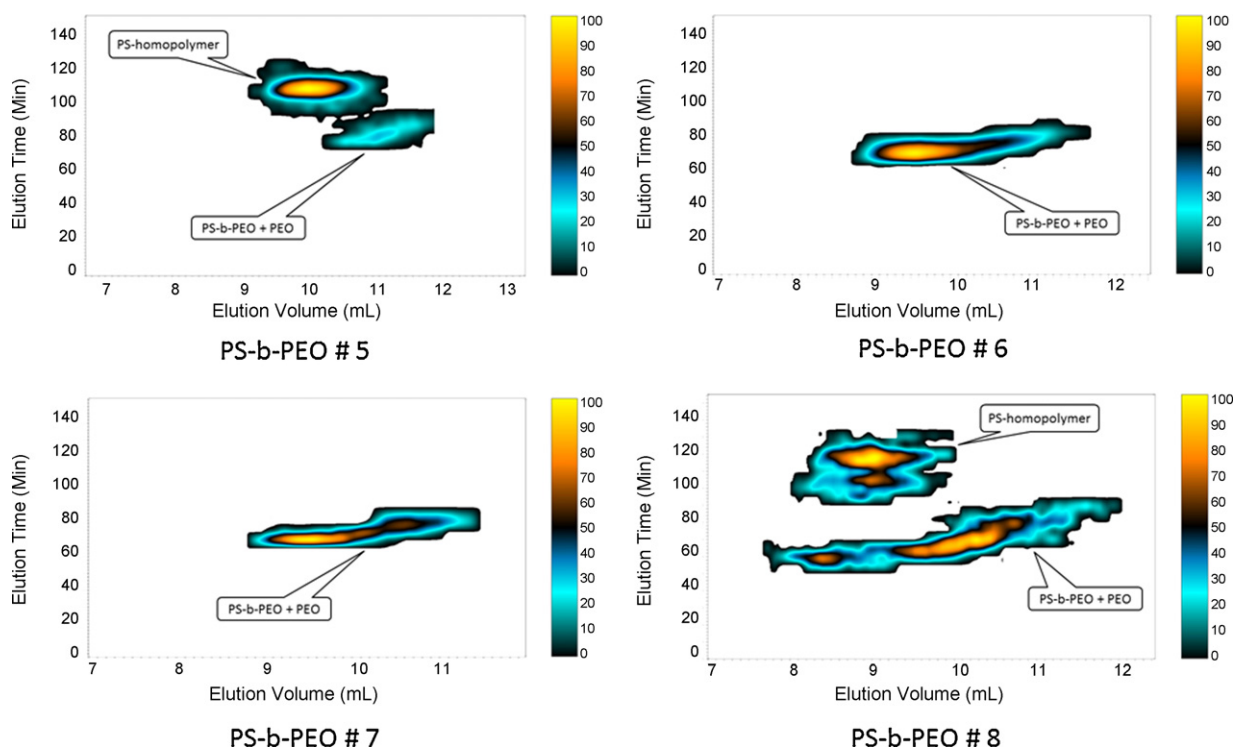


Fig. 5. Two-dimensional LC contour plots at the critical conditions of PS (Y-axis = min, X-axis = mL).

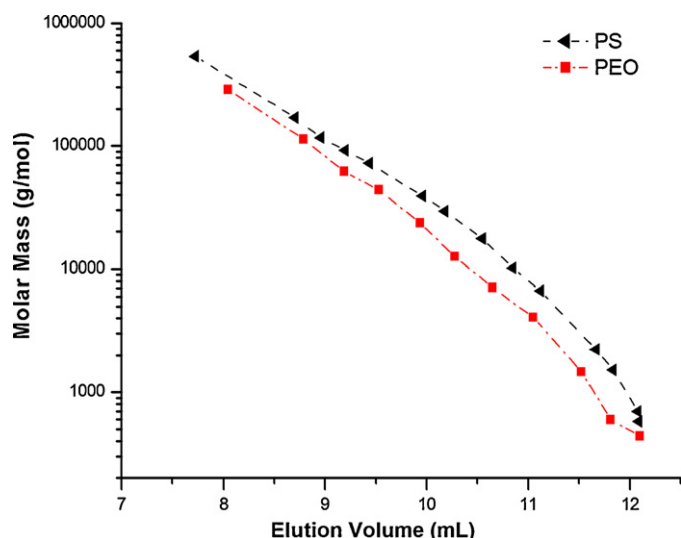


Fig. 6. SEC calibration curves of PEO and PS in the second dimension on a 50 mm × 20 mm PSS GRAM column in DMF.

from the rest of the sample. To confirm the absence of PS homopolymers from most of the above samples, we spiked one of the samples with a PS calibration standard and as can be seen it eluted at 3.30 mL (Fig. 3b). This confirms the absence of PS homopolymers in the samples shown in Fig. 3a (with sample PS-b-PEO # 1 being the exception). Fig. 4 shows fractionation of some of the products at these conditions. PS-b-PEO # 7 has a bimodal distribution but still does not contain any PS homopolymers as all the sample elutes earlier than 3.30 mL. On the other hand PS-b-PEO # 8 has a bimodal distribution and one of the major peaks is around the critical elution volume of PS.

Fig. 5 shows the two-dimensional contour plots of some of the products. The first-dimension separation is performed at the critical conditions of PS (Y-axis), while in the second dimension all separated fractions from the first dimension are analysed by online SEC (X-axis). PS-b-PEO # 5 and # 8 contain a significant amount of PS homopolymers which elute after 100 min in the first dimension (Y-axis). This fraction elutes earlier than the block copolymer fraction (along X-axis), indicating that the hydrodynamic volume of this fraction is greater than that of the fraction eluting before 100 min in the first dimension (Y-axis).

This suggests that they could be the coupling product of PS chains which are the first-synthesised block. Specifically, PS-b-PEO # 5 mainly consists of PS homopolymers and only a very small amount of block copolymers and/or PEO homopolymers. PS-b-PEO # 8 also contains a significant amount of PS homopolymers eluting earlier than the other fractions on the X-axis, indicating a larger hydrodynamic volume. The fraction eluting earlier than 100 min in the first dimension (Y-axis) has a very broad and multimodal distribution in the second dimension (X-axis), indicating differences in hydrodynamic volumes of different species (possibly block copolymers and PEO homopolymers). The other two products shown do not contain any PS homopolymers but have a bimodal distribution in the first dimension according to the PEO block and a very broad molar mass distribution. The hydrodynamic volumes of PSs and PEOs with similar molar masses are different from each other in DMF. This leads to different calibration curves for both types of polymers on the GRAM column used for SEC in the second dimension as shown in Fig. 6. This means that results regarding the molar mass distribution of the block copolymer are misleading. The PS calibration curve is used for getting an idea about the molar mass of the homopolymer fraction from the two-dimensional contour plots shown in Fig. 5. The molar mass information obtained is used

Table 2

The PS and PEO homopolymer content in the PEO-b-PS copolymer samples (deviations collected from three independent measurements).

PS-b-PEO #	PS homopolymers (%)		PEO homopolymers (%)	
	LCCC	FTIR	LCCC	FTIR
4	–	–	57 ± 3	–
5	34 ± 2	–	27 ± 2	–
6	–	–	48 ± 1	–
7	–	–	53 ± 3	57
8	32 ± 1	23	33 ± 3	45

to choose an appropriate calibration curve from Fig. 2b, for quantification of the PS homopolymers in the samples in one-dimensional analysis (Figs. 3a and 4). The results are summarised in Table 2.

As a next step, fractions of some of the products are collected and subjected to further analysis. As can be seen in Fig. 4, PS-b-PEO # 7 is fractionated into two fractions. Both fractions elute earlier than the critical elution volume of PS, so neither of them could be a PS homopolymer fraction. Both fractions are separated according to the PEO block length. PS-b-PEO # 8 is also separated into two fractions. Fraction-1 should contain the block copolymers and PEO homopolymers since it elutes earlier than the void volume of the column, while fraction-2 elutes at the CAP of PS and should contain only PS homopolymers. Fractions are collected manually according to the limits shown in Fig. 4. These fractions are subsequently analysed by LCCC of PEO, as well as by FTIR.

Fraction-1 of PS-b-PEO # 8 is separated from PS homopolymer but it might still contain PEO homopolymers along with real block copolymers. To confirm this, fractions are analysed by LCCC of PEO (refer Section 4.2). The elugrams of the raw product and the two fractions of PS-b-PEO # 8 (from critical conditions of PS, Fig. 4) are shown in Fig. 7 at the critical conditions of PEO. As can be seen, the raw product contains PEO homopolymer, which elutes at approximately 3.7 mL irrespective of molar mass, as does fraction-1 from PS-LCCC (Fig. 4). The polymers eluting earlier than the fraction at 3.7 mL are block copolymers, separated according to the molar mass of the PS block. Fraction-2 on the other hand does not contain any PEO homopolymers since this fraction was collected at the critical conditions of PS where it eluted at the void volume of the column.

These fractions were also analysed by FTIR. For qualitative FTIR, spectra were recorded in reflectance mode. Multiple spectra were recorded to ensure consistency of results. Fig. 8 illustrates the FTIR

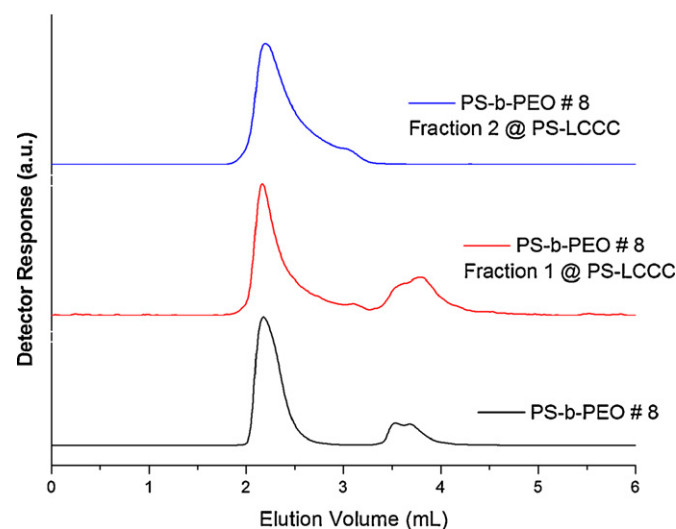


Fig. 7. PS-b-PEO # 8 raw product and its fractions from PS-LCCC (Fig. 4) analysed at the CAP of PEO on a Nucleosil Si 300 Å column in THF–DMF (96–4).

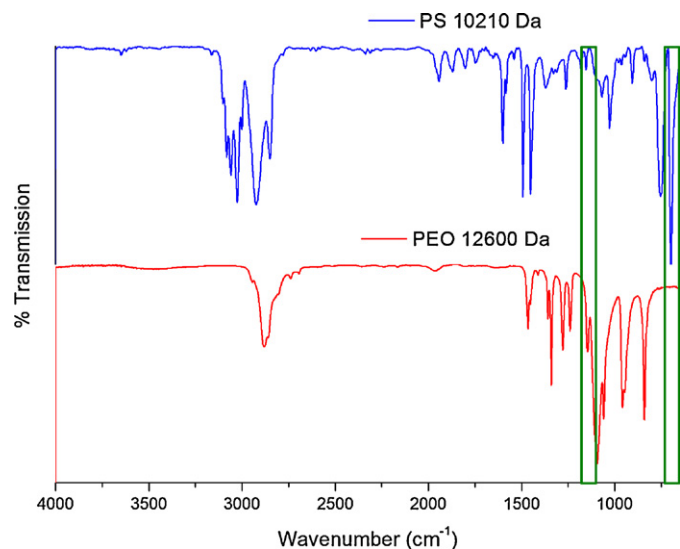


Fig. 8. FTIR spectra of PS and PEO homopolymer standards.

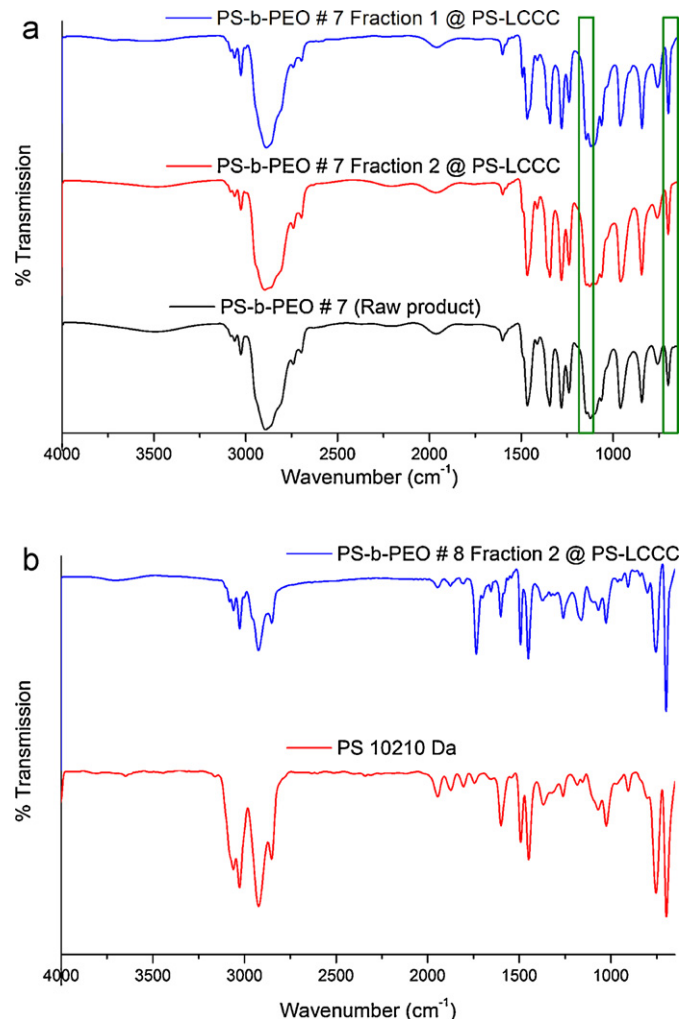


Fig. 9. (a) FTIR spectra of the PS-b-PEO # 7 raw product and its copolymer fractions from PS-LCCC (Fig. 4) and (b) FTIR spectra of the PS homopolymer fraction from PS-LCCC of PS-b-PEO # 8 (Fig. 4) and the corresponding PS homopolymer standard for comparison.

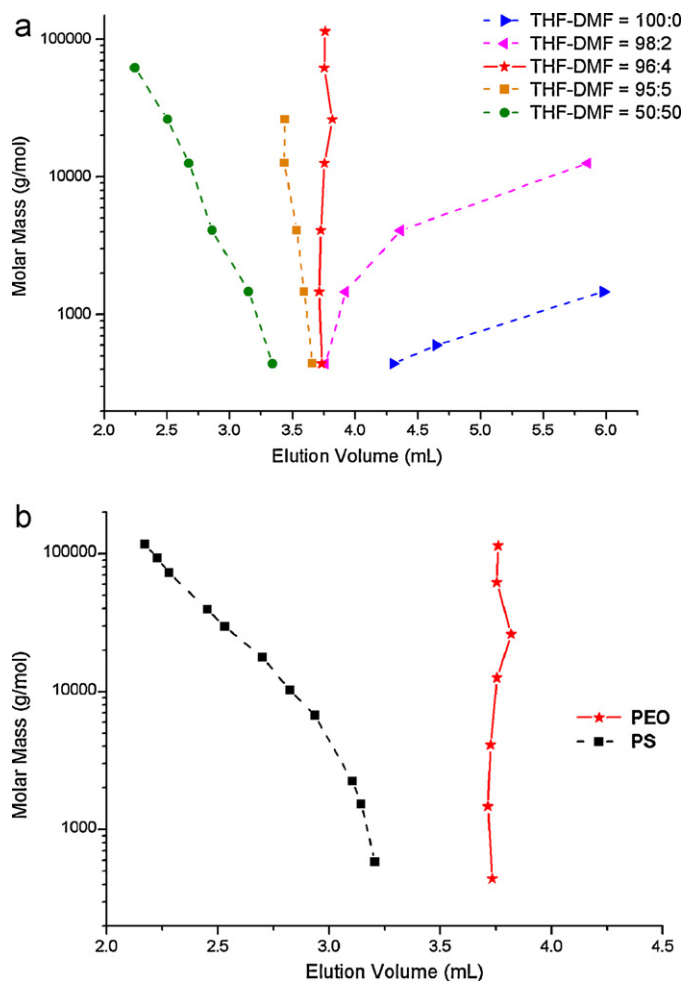


Fig. 10. (a) CAP of PEO on a Nucleosil Si 300 Å column in THF-DMF and (b) elution behaviour of PSs at the CAP of PEOs on a Nucleosil Si 300 Å column in THF-DMF (96–4).

spectra of homopolymer standards of each type of monomer. The different polymer types are clearly distinguishable from each other.

PS has a strong absorption at 700 cm^{-1} due to the asymmetric C–H stretching, while the strongest absorption of PEO is the triplet at 1100 cm^{-1} due to the C–O–C asymmetric stretching. The FTIR spectra of PS-b-PEO # 7 and its two fractions collected at critical conditions of PS (Fig. 4) are shown in Fig. 9a. The differences between some of the PS-b-PEO samples which have been highlighted by LCCC are further proven by FTIR analysis.

The two fractions of PS-b-PEO # 7 clearly contain both types of polymer (Fig. 9a). LCCC of PS has shown that this sample does not contain any PS homopolymers, however, the presence of PEO homopolymers in these fractions cannot be discounted and is in fact present (refer Section 4.2). PS-b-PEO # 8 on the other hand definitely contains PS homopolymer as demonstrated by LCCC. FTIR analysis (Fig. 9b) provides additional proof since it is evident that fraction-2 of PS-b-PEO # 8, collected at PS-LCCC, only contains PS homopolymer and there is no trace of PEO in the fraction. This confirms the successful separation of PS homopolymers from the rest of the sample at its critical conditions.

4.2. Separations at the CAP of PEO

To find a good solvent system which can dissolve both PEOs and PSs was a challenge since both polymers are incompatible with each other, as well as with good solvents for each other. The

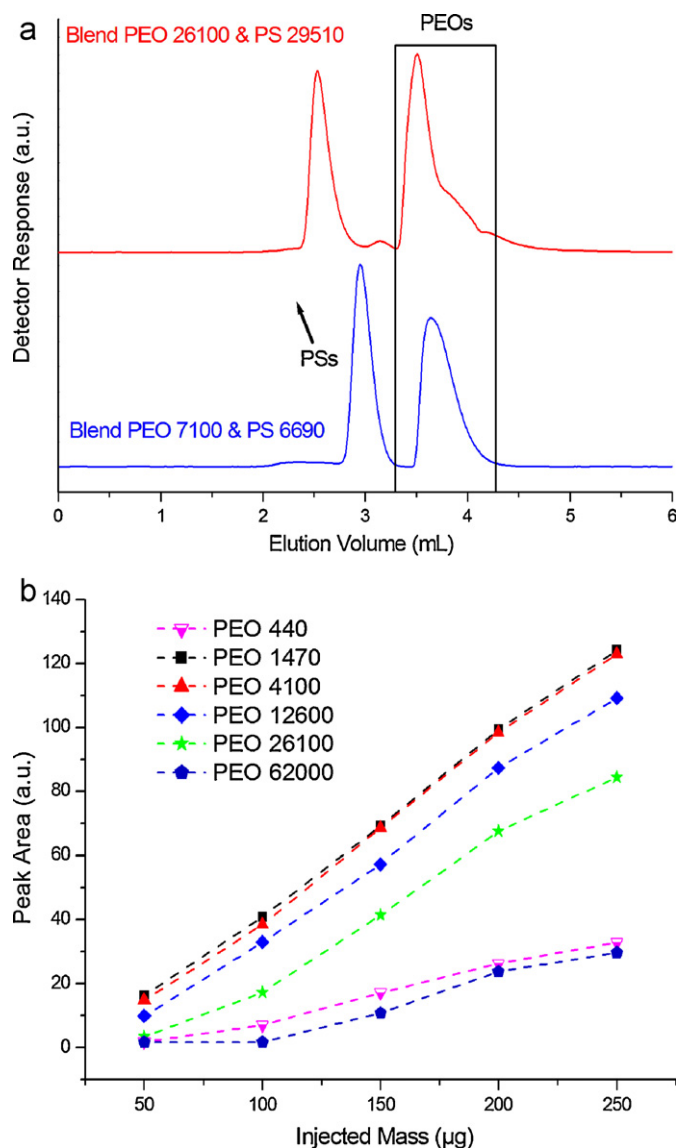


Fig. 11. (a) Elugrams of blends of PEO and PS calibration standards at the CAP of PEO on a Nucleosil Si 300 Å column in THF–DMF (96–4) and (b) ELSD calibration curves for PEO at its critical conditions (THF–DMF: 96–4) on a Nucleosil Si 300 Å column.

incompatibility increases dramatically with an increase in molar mass. The CAP of PEO was realised on a normal phase Si column in the same mobile phase system (THF–DMF) which is used for the critical conditions of PS. THF is an adsorption promoting solvent for PEOs on a normal phase Si column. The addition of DMF promotes desorption. At a mobile phase composition of 96–4 (THF–DMF, vol.%), all PEO standards elute at more or less the same elution volume and show no molar mass dependence. Further increase in the DMF content in the mobile phase leads to the exclusion of PEOs from pores of the stationary phase resulting in earlier elution of higher molar mass standards than the lower molar mass standards (Fig. 10a).

At the critical conditions of PEOs, PSs are excluded from the pores of the stationary phase and elute in order of decreasing molar mass (Fig. 10b). Fig. 11a depicts the blends of PEO and PS calibration standards with different molar masses. It can be seen that PEOs elute at the void volume of the column independent of their molar mass, while PSs show molar mass dependence and elute in the size exclusion regime earlier than the PEOs. The ELS detector is

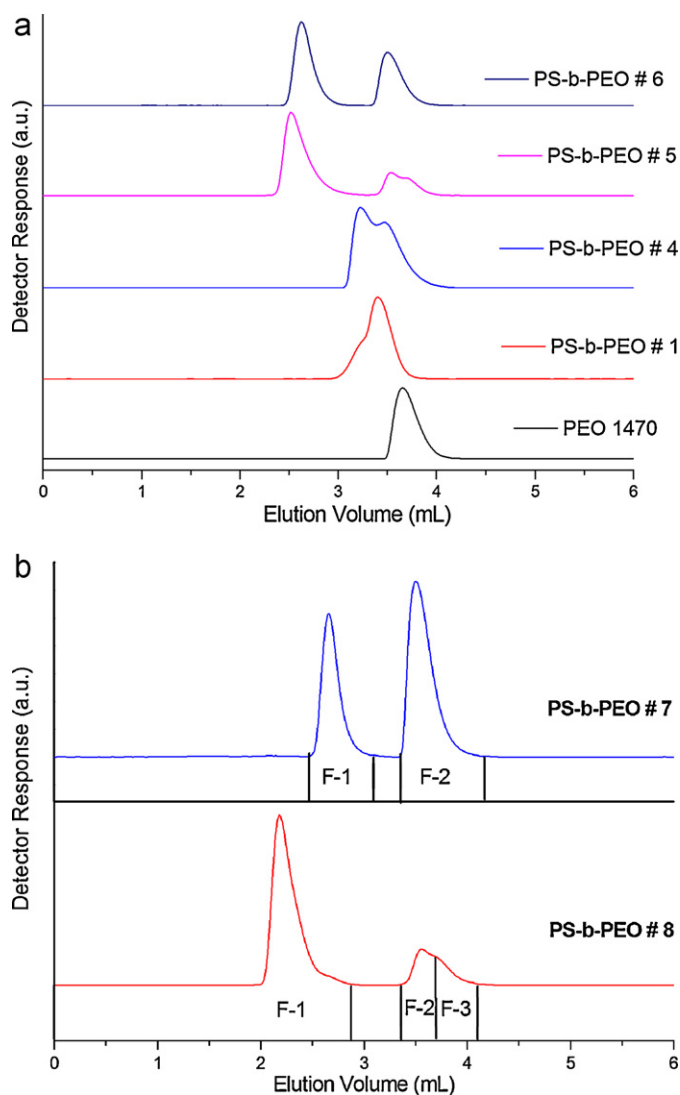


Fig. 12. (a) Elugrams of PS-b-PEO copolymers at the CAP of PEO on a Nucleosil Si 300 Å column in THF–DMF (96–4) and (b) fractionation of PS-b-PEO copolymers at the CAP of PEO on a Nucleosil Si 300 Å column in THF–DMF (96–4).

calibrated for PEOs by injecting varying masses of different molar mass standards. The ELSD calibration curves are given in Fig. 11b.

The reasons for the varying detector responses for different molar mass standards are explained earlier and the effects are more pronounced in this case. The molar mass information for PEO homopolymers from two-dimensional analysis of block copolymers (will be explained later, Fig. 13) was used to choose the appropriate calibration curve for quantification of PEO homopolymers in the samples.

These conditions can be used for the analysis of PS-b-PEO copolymers. The analysis method separates PEO homopolymers from block copolymers and PS homopolymers, and separates the block copolymers according to the molar mass distribution of the non-critical PS block. The determination of the molar mass distribution of the PS block is again not very reliable due to the same reasons as explained in Section 4.1. The elugrams of some of the products under study are shown in Fig. 12a and b. In contrast to the results obtained at the critical conditions of PS, where it was found that only two of the products contain an appreciable amount of PS homopolymers, almost all the products contain PEO homopolymers (eluting at approximately 3.7 mL).

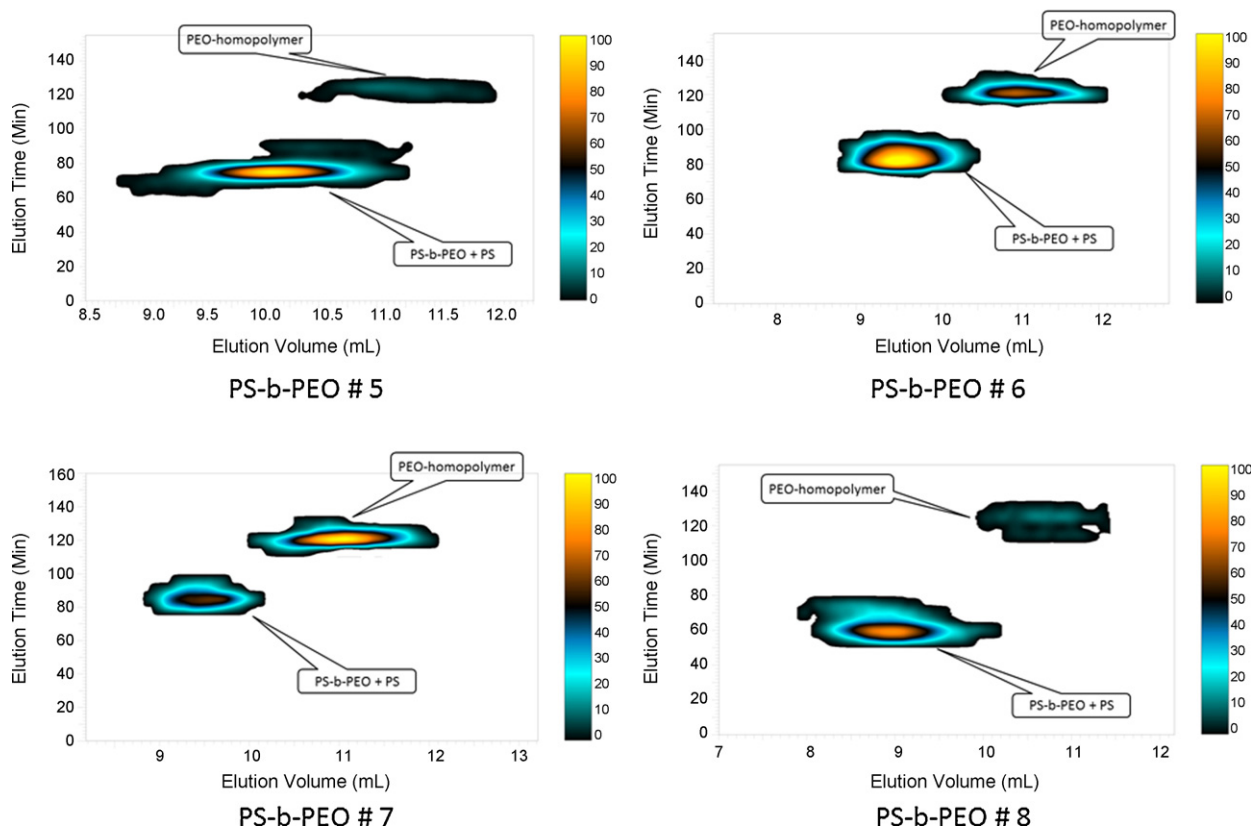


Fig. 13. Two-dimensional LC contour plots at the critical conditions of PEO (Y-axis = min, X-axis = mL).

The above-mentioned critical conditions for PEO are used as the first dimension for comprehensive two-dimensional liquid chromatography, with SEC being used as the second dimension. The Y-axis illustrates the elution time in the first dimension while the X-axis depicts the elution volume in the second dimension of all separated fractions from the first dimension (Fig. 13). All products shown contain a considerable amount of PEO homopolymers which elute at 120 min in the first dimension (Y-axis). The fraction eluting earlier than 120 min in the first dimension is a mixture of block copolymers and PS homopolymers. The calculation of molar mass distributions for block copolymers is again not reliable due to the same reasons as explained in the results at critical conditions of PS. The same methodology was followed for quantification of the PEO homopolymers content of the samples as was used for the quantification of the PS homopolymers content. The PEO calibration curve (Fig. 6) was used for determination of the molar mass distribution of the PEO homopolymer fraction. The molar mass information obtained was used to select the most appropriate calibration curve from Fig. 11b. The estimated amount of PEO homopolymers in the samples from LCCC analysis (Fig. 12a and b) is given in Table 2. The results are obtained by using the most appropriate calibration curve of a narrow molar mass calibration standard (Fig. 11b). It is worth mentioning at this point that the homopolymers formed during synthesis are neither narrow in molar mass distribution, nor have exactly the same molar mass as the calibration standards.

As a next step, fractions of some of the products are collected and subjected to further analysis at the CAP of PS, as well as FTIR, both qualitatively and quantitatively. The fractionation limits of some of the fractionated products are shown in Fig. 12b. PS-b-PEO # 7 contains a significant amount of PEO homopolymer and is collected as one copolymer fraction (fraction-1) and one homopolymer fraction (fraction-2), whereas PS-b-PEO # 8 is fractionated as one copolymer

fraction (fraction-1) and two homopolymer fractions (fraction-2 and fraction-3) due to its bimodal distribution.

These fractions are subjected to analysis at the critical conditions of PS. The PEO homopolymers are separated from the sample at its CAP but the block copolymer fraction might still contain PS homopolymers which elute with the copolymers. The elugrams of PS-b-PEO # 8 and its copolymer fraction (fraction-1), collected at the critical conditions of PEO (from Fig. 12b), are shown in Fig. 14 as analysed at the critical conditions of PS. It can be seen that the copolymer fraction contains significant amounts

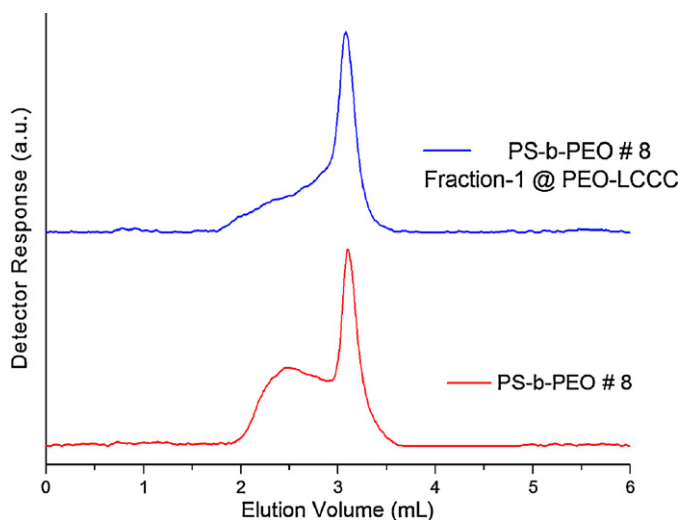


Fig. 14. PS-b-PEO # 8 raw product and its copolymer fraction from PEO-LCCC (Fig. 12b) analysed at the CAP of PS on a Symmetry RP octadecyl 300 Å column in THF-DMF (18–82).

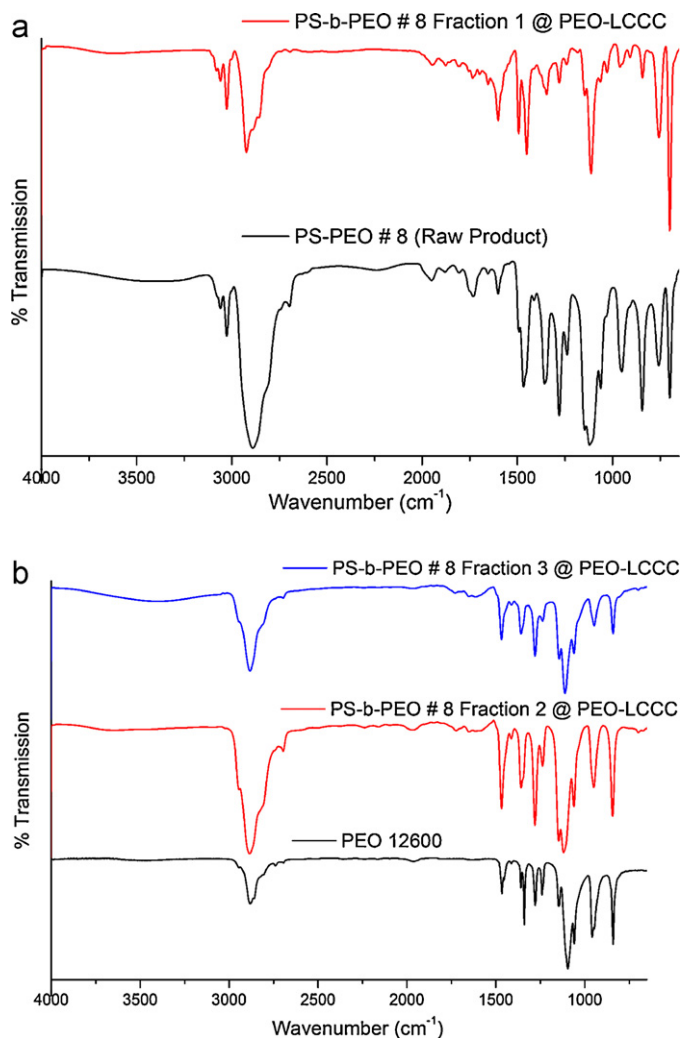


Fig. 15. (a) FTIR spectra of the PS-b-PEO # 8 raw product and its fraction-1, collected at the critical conditions of PEO (Fig. 12b) and (b) FTIR spectra of a PEO standard and the PEO homopolymer fractions (fractions 2 and 3) of PS-b-PEO # 8 collected at PEO-LCCC (Fig. 12b).

of PS homopolymers and that the real block copolymer is just the earlier eluting broad shoulder which has a very broad distribution according to the PEO block.

In order to confirm the separations obtained, the fractions collected at the critical conditions of PEO were analysed by FTIR. The spectra of both types of standards are given earlier (Fig. 8). Fig. 15a illustrates the FTIR spectra of the PS-b-PEO # 8 raw product along with its fraction-1, collected at the critical conditions of PEO (Fig. 12b). It is evident that fraction-1 contains both PS and PEO, although there appears to be a higher concentration of PS in the fraction as compared to the raw material. This is expected since the PEO homopolymers have been separated from this fraction. This is confirmed in Fig. 15b which shows that PS-b-PEO # 8 fraction-2 and fraction-3 contain only PEO homopolymers and that there is no trace of any PS in the fractions as was expected.

4.3. Quantification by solution FTIR

Quantification of the relative amounts of PS and PEO in the samples was accomplished by analysing the raw products and their copolymer fractions by transmission FTIR in solution. A calibration curve was constructed using PS and PEO standards (Fig. 16). The data are fitted with a 3rd order polynomial equation.

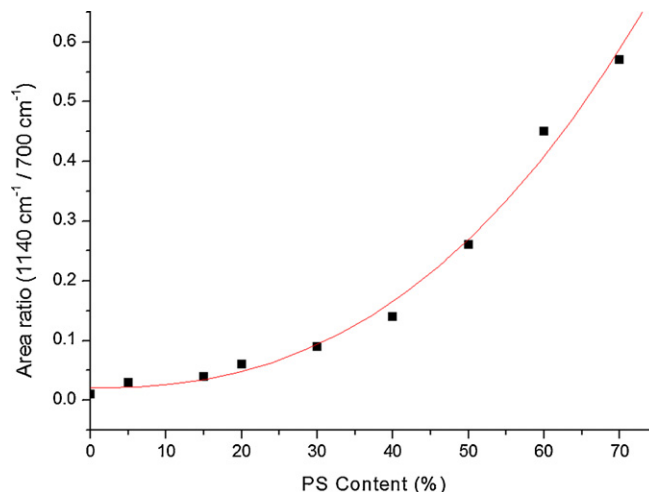


Fig. 16. FTIR calibration curve for PS/PEO content determination. The data are fitted with a 3rd order polynomial equation.

The chemical composition of the raw products as obtained by quantitative FTIR analysis by using the calibration curve is depicted in Fig. 17. The results for the chemical composition of the raw samples agree quite well with the information provided by the manufacturer (Table 1).

The fractions collected at both critical conditions were also analysed quantitatively by solution FTIR. Using an FTIR calibration curve, the amount of PS and PEO homopolymers in the samples are determined, the results of which are summarised in Table 2. The agreement of the results for the PS and PEO homopolymer quantification by LCCC and FTIR is fair. The results obtained by FTIR would be more reliable because the LCCC quantification is based on an appropriate calibration curve of narrow molar mass standards. The side products are neither narrow in molar mass, nor do they have exactly the same molar mass as the standards used. However, the calibration curves chosen are the most appropriate ones despite the fact that they can result in slight deviations from the real values.

The results obtained at critical conditions of PS and PEO are combined to obtain the overall composition of the products (Fig. 18a). The results obtained by the different approaches are in fair agreement. The chemical compositions of the real block copolymers in two of the products, as obtained by FTIR, are shown in Fig. 18b. The results show that the chemical composition of the real block copolymer is very different from that obtained for the raw samples.

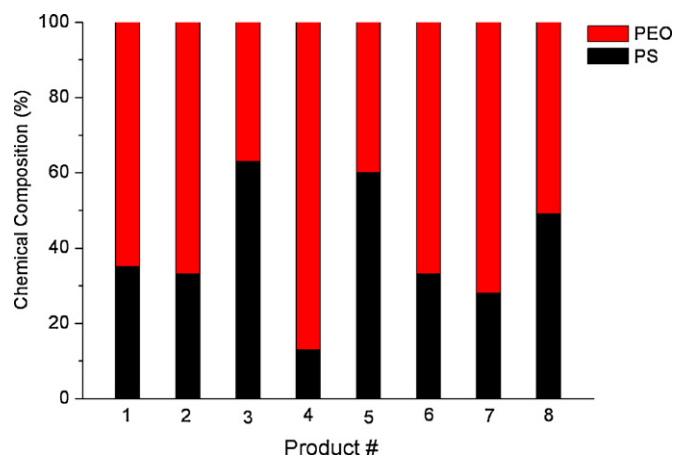


Fig. 17. Chemical composition of all products being investigated as obtained by quantitative FTIR analysis.

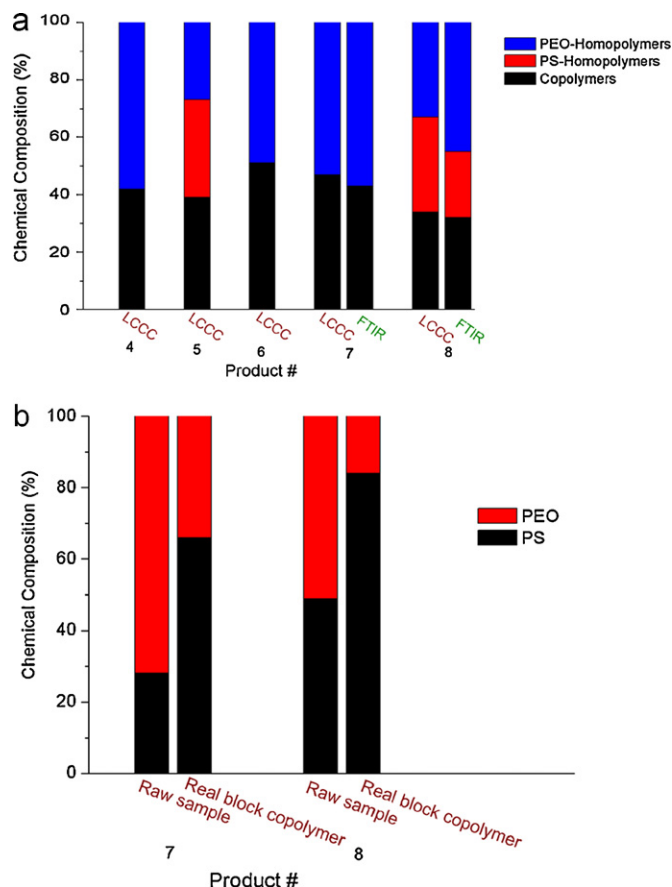


Fig. 18. (a) Comparison of the composition of the raw samples as revealed by LCCC and FTIR and (b) chemical composition of the real block copolymer as revealed by FTIR analysis.

5. Conclusions

Detailed analysis of PS-*b*-PEO copolymers at the critical conditions of PS and PEO showed that most of the products contain quite a large amount of both types of homopolymers formed during synthesis. The combination of comprehensive 2D-LC and the hyphenation of LCCC with LCCC of the other block, along with FTIR, gave much more insight into the product composition. The agreement of the quantification results for the homopolymers content, both by calibration of the ELS detector used for LCCC, and by FTIR is fair. The combination of the results at both critical conditions revealed the composition of the samples and the chemical composition of the real block copolymer, which constitutes less than 50% for almost all the samples. It is demonstrated that the results for the chemical composition of the raw samples, as obtained by FTIR analysis of the bulk products, can be misleading. The products under study are complex mixtures of homopolymers of both types along with the targeted block copolymers. The chemical compositions of the real block copolymers are very different from what is expected or obtained by direct spectroscopic analysis of the raw product.

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